

An Upper Boundary Condition Permitting Internal Gravity Wave Radiation in Numerical Mesoscale Models

JOSEPH B. KLEMP AND DALE R. DURRAN

National Center for Atmospheric Research,¹ Boulder, CO 80307

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ABSTRACT

A radiative upper boundary condition is proposed for numerical mesoscale models which allows vertically propagating internal gravity waves to pass out of the computational domain with minimal reflection. In this formulation, the pressure along the upper boundary is determined from the Fourier transform of the vertical velocity at that boundary. This boundary condition can easily be incorporated in a wide variety of models and requires little additional computation. The radiation boundary condition is derived from the linear, hydrostatic, Boussinesq equations of motion, neglecting Coriolis effects. However, tests of this radiation boundary condition in the presence of nonhydrostatic, Coriolis, nonlinear and non-Boussinesq effects suggest that it would be effective in many mesoscale modeling applications.

1. Introduction

Limited-area atmospheric prediction models are greatly complicated by the necessity of specifying a numerical domain which includes only a portion of a physically unbounded flow. Modelers are continually challenged with the problem of imposing boundary conditions which make the flow behave as it would if the boundaries were not there. In formulating conditions at an artificial domain boundary, two separate and often conflicting objectives are generally pursued. These are 1) to impose conditions from a larger scale flow field on the limited model domain, and 2) to allow internally generated disturbances to pass out through the boundaries without artificial reflection.

In recent years, considerable progress has been made toward the development of numerical boundary conditions which successfully satisfy these requirements at lateral boundaries. In contrast, the formulations used at the upper boundary are still rather primitive, particularly in their treatment of internal gravity waves. For scales of motion in the range of $O(10\text{--}1000)$ km, internal gravity waves may play an important role in transporting disturbances throughout the model domain. However, commonly used upper boundary conditions (free surface, rigid lid, etc.) are totally reflective to vertically propagating internal gravity waves, and the use of such conditions may produce significant errors in numerical simulations which resolve these scales of atmospheric

motion. For example, in cloud models, storms penetrating into the lower stratosphere may generate strong gravity wave motion near the top of the domain. If artificially reflected from the upper boundary, these waves may significantly distort the evolving storm structure. In larger scale mesoscale models, a reflective upper boundary may substantially alter the redistribution of momentum produced by gravity waves.

In models designed primarily to simulate gravity waves, such as mountain wave models (cf. Clark and Peltier, 1977; Klemm and Lilly, 1978), and critical layer models (Fritts, 1979), reflections from the upper boundary can completely distort the numerical solution. In order to minimize such reflections, these models include a viscous damping layer at the top of the computational domain to absorb vertically propagating waves before they reach the top of the numerical grid. Although absorbing layers can be effective, they are computationally inefficient, since a large portion of the total domain (often one-third to one-half) must be devoted to the absorbing region.

Radiation boundary conditions for finite domains have been derived for linear waves (Béland and Warn, 1975; Bennett, 1976). However, these formulations contain Laplace transforms whose evaluation requires the storage of the time history of the flow field at the boundary. This additional storage requirement can be unacceptable when numerical integrations are conducted over a large number of time steps.

The purpose of this paper is to present a boundary condition which radiates most of the gravity wave energy through the upper boundary and which is easy to incorporate in most numerical models. We believe

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that this formulation provides a significant improvement over the totally reflective boundary conditions currently employed in mesoscale models. In most applications it appears to be as effective as a viscous damping layer, and avoids the computational burden associated with the absorbing layer approach. In addition, it does not require the storage of boundary data which is necessary in other, more exact formulations.

We begin, in Section 2, by describing the fundamental differences between horizontal and vertical wave propagation which influence the formulation of a radiation condition. An exact radiation boundary condition which can be applied locally in time is then derived based on the linear, hydrostatic, Boussinesq equations of motion, neglecting Coriolis effects. Tests of this condition, presented in Section 3, for linear flow in a hydrostatic numerical mountain wave model demonstrate its stability, and show a high degree of accuracy when compared to analytic solutions. In Section 4, we evaluate the performance of this radiation condition over a wide variety of scales of motion (from nonhydrostatic to quasi-geostrophic), and examine the effects of nonlinearity and nonuniform atmospheric conditions. Although the radiation condition is only approximately correct under these more general conditions, it maintains a significant level of accuracy. Summary remarks are included in Section 5.

2. Basic concepts for a numerical radiation boundary condition

The wave-advection radiation boundary condition commonly used to transport gravity wave energy through lateral boundaries cannot be applied to the upper boundary because there are inherent differences between horizontal and vertical wave propagation. To illustrate this, we consider the simplest relevant set of equations which support gravity wave propagation: the linear, two-dimensional, hydrostatic, Boussinesq equations for a uniform mean state in the absence of Coriolis effects:

$$\frac{\partial u}{\partial t} + U \frac{\partial u}{\partial x} + \frac{\partial \phi}{\partial x} = 0, \quad (1)$$

$$\frac{\partial \theta}{\partial t} + U \frac{\partial \theta}{\partial x} + \frac{d\bar{\theta}}{dz} w = 0, \quad (2)$$

$$\frac{\partial \phi}{\partial z} - g \frac{\theta}{\theta_0} = 0, \quad (3)$$

$$\frac{\partial u}{\partial x} + \frac{\partial w}{\partial z} = 0. \quad (4)$$

Here (u, w) represent the perturbation velocity components in the Cartesian (x, z) coordinate system, and θ is the perturbation potential temperature. The background environment is characterized by a mean wind

U , a reference potential temperature θ_0 , and a mean potential temperature gradient $d\bar{\theta}/dz$. The heat capacity at constant pressure is c_p , g is the gravitational acceleration, and t is time. For the pressure variable, we define $\phi = c_p \theta_0 \pi$, with π being the perturbation Exner function. (The application of a radiation boundary condition to a more general set of equations is discussed in Section 4.)

The basic wave propagation characteristics for this system of equations are best illustrated by considering the dispersion equation produced by disturbances which are periodic in time and space such that

$$\xi(x, z, t) = \xi(k_x, l, \omega) \exp[i(k_x x + lz - \omega t)] \quad (5)$$

for each variable, where k_x and l represent the horizontal and vertical wavenumbers, and ω is the frequency. For disturbances of this form, the existence of a nontrivial homogeneous solution to (1)–(4) requires that

$$\omega = k_x(U \pm C), \quad (6)$$

where $C = N/|l|$ and $N^2 = (g/\theta_0)(d\bar{\theta}/dz)$. Here N is the Brunt-Väisälä frequency and C the intrinsic horizontal phase speed.

Horizontal gravity wave propagation is characterized by the horizontal phase speed,

$$C_{px} = \frac{\omega}{k_x} = U \pm C, \quad (7)$$

and the horizontal group velocity,

$$C_{gx} = \frac{\partial \omega}{\partial k_x} = U \pm C. \quad (8)$$

For each wavenumber pair (k_x, l) , the horizontal propagation speeds of the phase lines and the energy are the same. Consequently, outward propagating wave energy can be transmitted through a lateral boundary by numerically advecting disturbances out through the boundary based on their horizontal phase speeds. This procedure results in upstream differences for advection terms at the boundary which are numerically stable. Lateral radiation boundary conditions based on this concept have been implemented for a variety of applications with reasonable success (e.g., Pearson, 1974; Orlanski, 1976; Klemm and Lilly, 1978; Klemm and Wilhelmson, 1978; Clark, 1979).

The character of vertical wave propagation differs from horizontal propagation in certain significant respects which prohibit the application of a wave advection condition along an upper domain boundary. The vertical phase speed C_{pz} and group velocity C_{gz} are given by

$$C_{pz} = \frac{\omega}{l} = \frac{k_x}{l} (U \pm C), \quad (9)$$

$$C_{gz} = \frac{\partial \omega}{\partial l} = \mp \frac{k_x}{l} C. \quad (10)$$

Not only are these expressions different, but for $C > U$ the vertical phase speed and group velocity are in opposite directions! In this situation, disturbances would have to be appropriately advected into the domain from above in order to allow wave energy to pass upward through the boundary. Consequently, the implementation of a vertical wave advection boundary condition at the upper boundary would require the use of numerically unstable "downstream" differences whenever gravity waves are present with phase speeds greater than the mean wind.

Our objective is to specify a condition which will allow only gravity waves which propagate their energy upward to exist at the upper boundary. In order to identify these wave modes, we consider linear disturbances of the form²

$$\xi(x, z, t) = \hat{\xi}(k_x, z, \omega) \exp[i(k_x x - \omega t)], \quad (11)$$

which, when substituted into (1)–(4), yield the familiar wave equation

$$\frac{\partial^2 \hat{w}}{\partial z^2} + \frac{N^2}{(U - C_{px})^2} \hat{w} = 0 \quad (12)$$

for an atmosphere having constant mean wind and stability. Solutions to (12) describe the vertically propagating gravity wave modes, and may be written

$$\hat{w} = A \exp\{i \operatorname{sgn}(k_x) N z / (U - C_{px})\} + B \exp\{-i \operatorname{sgn}(k_x) N z / (U - C_{px})\}. \quad (13)$$

As established by Eliassen and Palm (1960), the vertical energy transport associated with these modes is given by

$$E_H = \rho_0 \overline{\phi w} = \frac{\rho_0 N}{2|k_x|} \{|A|^2 - |B|^2\}. \quad (14)$$

Thus the term in (13) with coefficient A corresponds to upward energy transport, while the term with coefficient B produces downward transport. The corresponding vertical momentum flux may be either negative or positive depending on the sign of $U - C_{px}$.

Wave reflection at the upper boundary, $z = z_t$, is suppressed if $B = 0$ in (13) for all propagating modes. This condition is satisfied by requiring

$$\frac{\partial \hat{w}}{\partial z} = \frac{i \operatorname{sgn}(k_x) N}{U - C_{px}} \hat{w}. \quad (15)$$

Transforming variables according to (11), Eqs. (1) and (4) can be combined such that

$$\frac{\partial \hat{w}}{\partial z} = -ik_x \hat{u} = \frac{ik_x}{U - C_{px}} \hat{\phi}. \quad (16)$$

Equating the right-hand sides of (15) and (16) then yields

$$\hat{\phi} = \frac{N}{|k_x|} \hat{w} \quad \text{at} \quad z = z_t. \quad (17)$$

Since the coefficient multiplying w in (17) has no frequency dependence, the time transformation in (17) can be removed with the result that

$$\phi_k(k_x, z_t, t) = \frac{N}{|k_x|} w_k(k_x, z_t, t), \quad (18)$$

where

$$(\phi_k, w_k) = (\phi, w) e^{-ik_x x} \quad (19)$$

are variables which have been transformed in the x direction only. Eq. (18) provides the desired upper boundary condition for pressure ϕ in terms of vertical velocity.³ In a stable atmosphere, this condition ensures an upward energy flux in (14) since the correlation between w_k and ϕ_k in (18) is always positive. [If the perturbation pressure p were used in (1) and (3) instead of ϕ , $p_k/\rho(z_t)$ would replace ϕ_k in (18).]

The radiation condition (18) is particularly well suited for use in time-dependent numerical models since it depends only on the instantaneous features of the flow at the boundary. Radiation conditions derived for more general governing equations than (1)–(4) contain time integrals, and their application in numerical models requires storage of the past history of the flow along the boundary (Béland and Warn, 1975; Bennett, 1976). Even in the hydrostatic, irrotational equations (1)–(4), the conventional form of the radiation condition (15) depends on the phase speeds (frequencies) of the individual modes, and thus would require the evaluation of integrals over time at the boundary. In contrast, the relationship between the transformed pressure and vertical velocity in (17) is fortuitously independent of frequency, allowing transient disturbances to be correctly radiated by a condition (18) which is applied locally in time.

The numerical implementation of (18) is quite simple. After computing the vertical velocity at a new time level, w is Fourier transformed horizontally at $z = z_t$, and each Fourier coefficient is multiplied by $N/|k_x|$. The pressure at the new time level along the upper boundary is then recovered through the reverse Fourier transform. Since w is real, the positive and negative wavenumber Fourier coefficients of w are complex conjugates, so only the calculation of the positive wavenumber coefficients is required when evaluating (18). The additional computations re-

² If temporal decomposition is actually required in evaluating a radiation condition, Laplace transforms should be applied to the initial value problem specified for model integrations. However, since we shall remove this temporal decomposition from the final result, we represent the time variation with a single Fourier mode to simplify the presentation.

³ This radiation boundary condition was proposed in a paper presented by Dr. Klemp at the AMS Third Conference on Atmospheric Waves and Stability, San Diego, January 1981.

quired to apply this boundary condition have minimal impact on overall model efficiency.

Applying this Fourier transform procedure in a finite domain implies a periodicity of disturbances at the upper boundary. The radiation condition (18) can also be expressed in physical coordinates by applying the convolution theorem to (18), with the result

$$\phi(x, z_t, t) = \frac{N}{\pi} \int_{-\infty}^{\infty} w(x', z_t, t) \ln|x - x'| dx'. \quad (20)$$

where $\pi = 3.1416$. Evaluation of this integral along the top of a domain which has finite horizontal dimensions would require values of $w(x, z_t, t)$ outside the domain. Thus, whether or not periodicity is assumed in formulating a radiation upper boundary condition, some flow structure must be specified outside the lateral extent of the computational domain. This complication is also present in the radiation condition derived by Beland and Warn (1975) for more general governing equations. Bennett (1976) proposed a technique for evaluating a radiation condition which requires only data along the finite boundaries, though again, storage of the time history of the boundary data is required. If the dominant disturbances are contained primarily in the central portion of the domain, the implied periodicity of our procedure will not adversely affect its radiative features. Since our goal is to devise an approximate condition which radiates most of the wave energy and is easy to apply in numerical models, we feel the assumption of periodicity is justified. For certain applications, some improvement might be achieved by artificially extending the vertical velocity field at the upper boundary and then transforming along this expanded surface.

In deriving the radiation boundary condition, the assumption of two-dimensionality is easily removed. By considering the three-dimensional equivalent of (1)–(4) with a mean wind component V in the y direction, Eq. (18) becomes

$$\phi_k = \frac{Nw_k}{(k_x^2 + k_y^2)^{1/2}}, \quad \text{at } z = z_t, \quad (21)$$

where k_y is the wavenumber in the y direction. This expression is identical to that in two dimensions, except that it now requires a two-dimensional Fourier transform along the upper surface of the domain. The derivation of (21) is discussed further in Section 4.

The radiation condition, proposed above, has been derived under a highly simplified set of assumptions; it is strictly valid only for linear, hydrostatic, Boussinesq flow in the absence of Coriolis effects. The mean state conditions are assumed to be horizontally homogeneous at the top of the computational domain, and a periodicity of disturbances at the upper boundary is implied by the Fourier transform procedure in a finite domain. Clearly, all of these as-

sumptions may be significantly violated in realistic mesoscale model simulations. However, as will be described in Section 4, tests indicate that this radiation condition remains effective over a wide range of conditions and appears to be suitable for capturing the essence of gravity wave radiation for many modeling applications on the meso- β and meso- γ scales.

3. Numerical application of the radiation boundary condition

Before considering more general equations, we shall demonstrate the stability and accuracy of the proposed radiation upper boundary condition in numerical solutions to (1)–(4). For numerical testing purposes, we shall consider the evolution of mountain waves in stable air flowing over a mountain contour. The mountain wave structures are particularly sensitive to the vertical transport of wave energy and both linear and nonlinear analytic solutions are available for comparison with model results. We emphasize, however, that the radiation condition is not designed specifically for the mountain wave problem; this application only illustrates the influence of the upper boundary condition on internal gravity waves in a numerical model.

The mountain wave model used for this initial testing was adapted from the two-dimensional model described by Durran (1981) and Durran and Klemp (1982). Although the original model is nonlinear, nonhydrostatic and compressible, we have altered it to be linear, hydrostatic and Boussinesq, so that the model equations correspond exactly to (1)–(4). The model marches the equations forward in time using leapfrog time differencing on a staggered grid with second-order spatial differencing, except for the differencing of the horizontal advection terms which is fourth order. The model incorporates a transformed vertical coordinate which conforms to the terrain along the lower boundary of the model domain. It also contains open lateral boundary conditions and an optional upper absorbing layer which is used for comparisons with the radiation condition. For further model details, the reader is referred to Durran and Klemp (1982).

The radiation upper boundary condition (18) is easily incorporated into the numerical procedures of any model which integrates (1)–(4). After stepping the horizontal momentum (1) and thermodynamic (2) equations forward to the new time level, $t + \Delta t$, the continuity equation (4) is vertically integrated (using a lower boundary condition for w) to obtain w along the top of the domain. The pressure at the top of the domain is then calculated by Fourier transforming w along the upper boundary, applying (18) to each Fourier coefficient, and inverse transforming to retrieve $\phi(x, z_t, t + \Delta t)$. The pressure is then computed by integrating the hydrostatic equation (3)

downward using $\phi(x, z_i, t + \Delta t)$ as the upper boundary condition. This completes the calculations for the new time step. In numerically integrating (1)–(4), no other boundary conditions are required at the upper boundary of the domain. For more general governing equations, other conditions must be specified; these are discussed in Section 4b.

In applying the radiation boundary condition to the finite difference equations of a numerical model, the accuracy and stability of the boundary condition can be somewhat improved by using the equivalent form of (18) derived from the finite-difference equations. The derivation and analysis of this numerical radiation condition is discussed in the Appendix. It requires only a minor adjustment in which k_x is replaced by its finite-difference form based on the differencing of the horizontal pressure gradient term in the momentum equation (see Appendix for details). Although we continue to refer to (18) in the discussion of its testing in numerical models, the form (A3) is actually being used.

The maximum allowable time step permitted in the model simulations using the radiation upper boundary condition is about the same as that attained in equivalent simulations using $\phi = 0$ as the upper boundary condition. Since the pressure along the radiating upper boundary responds to internally generated disturbances, it is more sensitive to numerical fluctuations. Consequently, in changing from a constant pressure upper boundary condition to the radiation condition, the time splitting tendency in the leapfrog differencing is increased. However, the standard Robert (1966) time filter, with a smoothing coefficient of 0.1 applied throughout the domain, appears to satisfactorily control this weak instability. Bougeault (1983) has proposed an implicit procedure for implementing this radiation condition which removes this splitting tendency.

For the model simulations, we consider solutions for flow of a uniform atmosphere (U and N constant) over a bell-shaped mountain contour, i.e.,

$$h(x) = \frac{H}{1 + (x/a)^2}. \quad (22)$$

The finite difference increments are chosen such that $\Delta x = 0.2a$, $\Delta z = 0.04\lambda_z$ and $\Delta t = 0.01a/U$, where $\lambda_z = 2\pi U/N$ is the steady-state vertical wavelength. The top of the model domain is set at $z_t = \frac{3}{4}\lambda_z$ while the lateral boundaries are located at $x = \pm 9a$. To avoid large artificial disturbances during start up, the horizontal wind speed is gradually increased from zero to its mean value over a dimensionless time of $Ut/a = 5$.

The nearly steady-state streamlines from the numerical solution at $Ut/a = 60$ are shown in Fig. 1 (displayed for a mountain amplitude of $NH/U = 0.5$) along with the steady-state linear analytic solution (see Alaka, 1960). We believe the small differences,

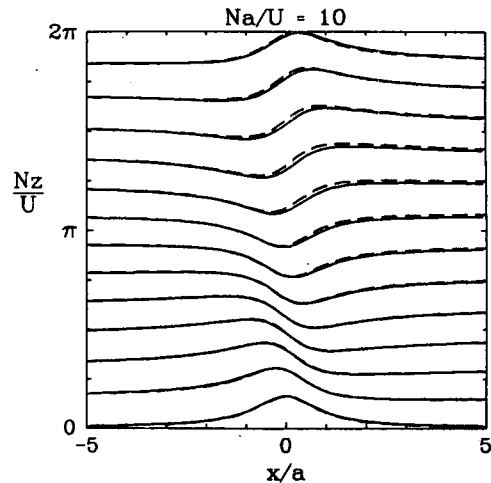


FIG. 1. Linear, hydrostatic Boussinesq streamlines for uniform flow over a bell-shaped mountain with $z_t = \frac{3}{4}\lambda_z$, displayed for $NH/U = 0.5$. Solid lines denote the numerical solution with the upper radiation condition at $Ut/a = 60$ and dashed lines represent the steady linear analytic solution.

apparent in Fig. 1, are not produced by the upper boundary condition because a second numerical solution (not shown), computed using a wave absorbing layer of thickness $1.5\lambda_z$ at the top of the domain, is virtually identical to the one generated with the radiation boundary condition. Klemp and Lilly (1978) have shown that such an absorbing layer can eliminate the spurious reflection of linear gravity waves very effectively. The wave absorber was constructed by adding Rayleigh damping terms to (1) and (2) which increase gradually in magnitude with height [see Durran (1981) for details].

The time-dependent evolution of this mountain wave flow is illustrated in Fig. 2 by the vertical profiles of the momentum flux, $M = \rho_0 u w$, obtained using either the radiation condition or an absorbing layer, at various times during each simulation. These profiles are normalized by the steady-state linear hydrostatic momentum flux from the analytic solution, given by

$$M_H = \frac{-\pi}{4} \rho_0 N U H^2. \quad (23)$$

Again, the two solutions exhibit quite similar characteristics. These flux profiles develop somewhat more slowly than the analytic time-dependent profiles derived by Klemp and Lilly (1980) because of the gradual start-up of the mean wind in the model.

To further evaluate the vertical radiation of propagating or transient gravity waves, we have recomputed the mountain wave evolution by expanding the horizontal domain, setting the mean wind speed to zero, and then towing the mountain contour along the lower boundary at a speed $-U$. Although the physical problem is unaltered by this coordinate

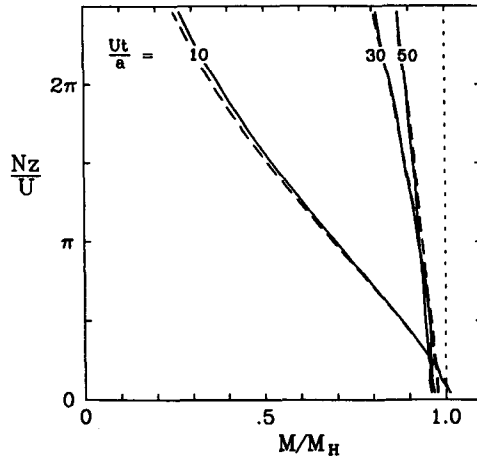


FIG. 2. Momentum flux profiles at $Ut/a = 10, 30, 50$ for simulations described in Fig. 1 obtained with the numerical radiation condition (solid lines) and with the numerical absorbing layer (dashed lines).

transformation, the waves are propagating horizontally ($C_{px} = -U$) in the framework in which the radiation condition is applied. The results from this simulation are indistinguishable from those displayed in Figs. 1 and 2 for the stationary mountain.

4. Radiation characteristics in more general applications

Since the radiation condition (18) is exact for (1)–(4), the simplified model equations tested above should exactly reproduce the analytic radiating solution, except for small numerical discrepancies. The results of the previous section demonstrate this fact, and confirm the numerical stability of the radiation boundary condition. However, to be generally useful, (18) must also allow the radiation of gravity waves in a wide variety of more realistic conditions. In this section we will demonstrate that the radiation boundary condition may remain effective in the presence of nonhydrostatic effects, Coriolis effects, nonlinear effects, mean density variation, and nonuniform atmospheric conditions.

The ability of this boundary condition to radiate linear waves in more general physical situations will be evaluated through quasi-analytic solutions of various forms of the linear, nonhydrostatic, anelastic equations which include Coriolis effects:

$$\frac{d_H u}{dt} + \frac{\partial \phi}{\partial x} = f v, \quad (24)$$

$$\frac{d_H v}{dt} + \frac{\partial \phi}{\partial y} = -f u, \quad (25)$$

$$\frac{d_H w}{dt} + \frac{\partial \phi}{\partial z} = g \frac{\theta}{\theta_0}, \quad (26)$$

$$\frac{d_H \theta}{dt} + \frac{d\theta}{dz} w = 0, \quad (27)$$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} - \beta w = 0, \quad (28)$$

where

$$\frac{d_H}{dt} = \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} + V \frac{\partial}{\partial y}, \quad (29)$$

β is the inverse density scale height, $\beta = -d \ln \bar{\rho} / dz$, and f is the Coriolis parameter. Emphasis will be placed on simplified analyses which provide a qualitative understanding of the accuracy to be expected under a variety of conditions. Nonlinear effects will be tested with numerical models by comparing results with simulations using an upper absorbing layer, and with solutions to Long's equation for hydrostatic flow.

a. Nonhydrostatic and rotational effects

The correct linear radiation condition becomes significantly more complicated when nonhydrostatic or Coriolis influences are present. Consider the Bousinesq ($\beta = 0$) form of (24)–(28) for periodic disturbances given by the three-dimensional equivalent of (11). Following the same development described in Section 2, we obtain the more general radiation condition:

$$\hat{\phi} = \frac{N}{(k_x^2 + k_y^2)^{1/2}} \left\{ 1 - \frac{[k_x U + k_y V - \omega]^2}{N^2} \right\}^{1/2} \times \left\{ 1 - \frac{f^2}{[k_x U + k_y V - \omega]^2} \right\}^{1/2} \hat{w}. \quad (30)$$

The influences of nonhydrostatic and rotational effects appear in the first and second terms in braces, respectively, and permit the vertical radiation of wave energy only for $f^2 < (k_x U + k_y V - \omega)^2 < N^2$. These correction terms represent the factors by which the momentum flux associated with each wavenumber pair is reduced from its hydrostatic nonrotational value.

For nearly stationary disturbances, (30) provides a viable boundary condition by assuming $\omega = 0$, and using the appropriate evanescent condition when either of the square root terms becomes imaginary. However, for propagating disturbances, (30) becomes ambiguous since the frequencies associated with each wavenumber pair cannot readily be determined in most numerical models. The application of (30) in a numerical model would require the evaluation of Laplace transforms for which the past history of the flow at the boundary must be stored (Bennett, 1976).

In seeking a boundary condition which radiates most of the transient and steady internal gravity wave energy and which is easy to apply in numerical mod-

els, we propose that these correction terms in (30) be ignored. With this approximation, the time transformation can be removed and (30) reduces to (21). In model applications which resolve horizontal scales allowing vertically propagating gravity waves, we will show that this simplified radiation condition may remain quite effective. This condition is clearly not suitable for radiating the much larger scale Rossby waves.

The radiation condition (18) or (21) can be easily applied in nonhydrostatic numerical models which use either the anelastic or compressible versions of the equations. In both approaches, the pressure along the top of the model domain is obtained directly from (18) or (21), using the same forward and reverse Fourier transform procedure as described in Section 3 for hydrostatic models. In anelastic models, pressure is obtained from the solution of a Poisson equation, and the radiation condition provides the required upper boundary condition for that equation. If the equations are expressed in terms of the streamfunction and vorticity, an upper boundary condition is required for the streamfunction. A convenient form of the approximate radiation condition is achieved by differentiating (18) with respect to z and then using $w_k = -ik_x \psi_k$ and the transformed hydrostatic equation (3), which yields

$$\left. \frac{\partial \psi_k}{\partial z} \right|_{z=z_t} = i \operatorname{sgn}(k_x) \frac{g}{N\theta_0} \theta_k(k_x, z_t, t). \quad (31)$$

This expression provides a Neumann type boundary condition which is used in solving the Poisson equation for the streamfunction.

In the compressible equations of motion, the continuity equation (28) becomes a prognostic equation for density (or pressure), and solution of a Poisson equation is no longer required. For these equations, the pressure obtained from the radiation condition is used to evaluate the finite-difference representation of the vertical pressure gradient in the vertical momentum equation (26), at the uppermost grid level.

To demonstrate the accuracy of the approximate radiation condition when nonhydrostatic or Coriolis effects are present, we again consider the two-dimensional, Boussinesq, steady flow of a uniform atmosphere past a bell-shaped mountain [Eq. (22)] with arbitrary half-width a , and compare the analytic representations for the steady solution derived using either (18) or the correct radiation condition (30) with $\omega = 0$. Notice that even at steady state, the terms in braces in (30) do not become unity; in fact, the magnitude of their deviation from unity at steady state is comparable to that during the transient response (due to the presence of a mean wind). Consequently, evaluation of the accuracy of the approximate radiation condition for the steady-state waves also provides an indication of the accuracy to be expected during the transient development.

Under the assumptions mentioned above, the solution to (24)–(28), subject to the approximate boundary condition (18) at $z = z_t$, is

$$\delta_k = \pi a H \exp[-a|k_x|] \times \frac{\alpha \cos l(z - z_t) + i \operatorname{sgn}(k_x) \sin l(z - z_t)}{\alpha \cos l z_t - i \operatorname{sgn}(k_x) \sin l z_t}. \quad (32)$$

where

$$l^2 = \frac{N^2}{U^2} \left(1 - \frac{k_x^2 U^2}{N^2} \right) \left(1 - \frac{f^2}{k_x^2 U^2} \right)^{-1}, \quad (33)$$

$$\alpha^2 = \left(1 - \frac{k_x^2 U^2}{N^2} \right) \left(1 - \frac{f^2}{k_x^2 U^2} \right). \quad (34)$$

Here δ is the vertical displacement of a streamline from its undisturbed level, and δ_k denotes its horizontal Fourier transform.

The vertical flux of horizontal momentum is a particularly sensitive measure of the radiative properties of the solution. Including the influences of rotation (Jones, 1967; Eliassen, 1968), it is defined by

$$M = \rho_0 \int_{-\infty}^{\infty} (u - f\eta) w dx, \quad (35)$$

where η is the displacement in the y direction. Using the relationships $U^2 \partial^2 \eta / \partial x^2 = -fu$ and $u = -U \delta \delta / \delta z$, the momentum flux becomes

$$M = -\frac{\rho_0 U}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(1 - \frac{f^2}{k_x^2 U^2} \right) \times \frac{\partial \delta_k}{\partial z} \exp(ik_x x) dk_x \cdot w dx. \quad (36)$$

Using (32) to evaluate (36) yields the momentum flux produced by the approximate radiation condition:

$$\frac{M}{M_H} = \frac{16a^3}{\pi} \int_0^{\infty} \int_0^{\infty} \frac{\alpha^2 e^{-ak_x} \sin k_x x dk_x}{\sin^2 l z_t + \alpha^2 \cos^2 l z_t} \cdot \frac{x dx}{(x^2 + a^2)^2}, \quad (37)$$

where M_H is the steady-state linear hydrostatic momentum flux given by (23). The momentum flux corresponding to the correct radiation condition is recovered by replacing α^2 with α in the numerator of (37), setting $\alpha^2 = 1$ in the denominator, and integrating over k_x only between f/U and N/U [Eq. (37) in Klemp and Lilly (1980) except that the e^{-ak_x} factor was omitted through typographical error]. Notice that in (37) the momentum flux now also depends on the height z_t at which the radiation condition is applied.

The momentum flux has been numerically evaluated from (37) as a function of horizontal scale (Na/U) for various values of Nz_t/U . These results are displayed in Fig. 3 for the case of $f/N = 10^{-2}$. The horizontal mode containing the maximum energy, $k_m = a^{-1}$, is vertically propagating only over the range $1 < Na/U < 100$; however, since the mountain profile

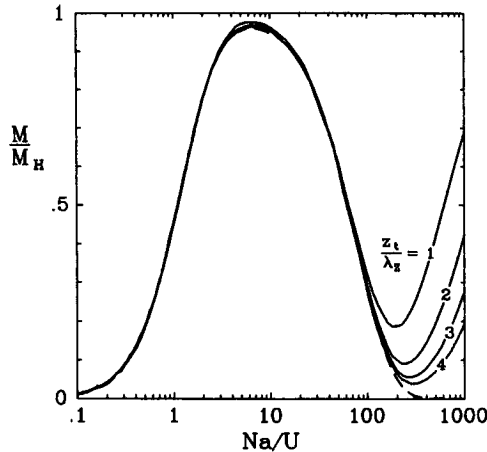


FIG. 3. Momentum flux for steady linear Boussinesq flow over a bell-shaped mountain with nonhydrostatic and Coriolis effects included for $f/N = 10^{-2}$. The momentum flux associated with the approximate radiation condition (18) is shown in solid lines for $z_i/\lambda_z = 1, 2, 3, 4$ as labeled in the figure. The momentum flux produced by the correct radiation condition is included with a dashed line.

(22) forces a broad spectrum of wavenumbers, the momentum flux for the correct radiation condition decreases more gradually as Na/U increases toward 100 or decreases toward unity. The vertical momentum flux produced with the approximate radiation condition (18) corresponds closely to the correct values throughout the main range of scales contributing to vertical energy propagation. Good agreement also occurs throughout the smaller scale, nonhydrostatic range where vertical transport is strongly curtailed.

As Na/U approaches 10^3 in magnitude, the momentum flux produced by the approximate condition (18) artificially increases, and in the limit of quasi-geostrophic flow, M/M_H approaches unity. At these large scales, disturbances decay only very gradually with height; thus spurious reflections at the upper boundary have greater influence in distorting the solution. Other boundary conditions, such as a rigid lid or a free surface condition, produce comparable errors in the flow field, though they do not contribute to a finite vertical momentum flux. These stationary large-scale modes are of little interest in modeling applications owing to the long time scale associated with their formation. For example, when $Na/U = 10^3$, more than 100 days would be required to produce nearly steady waves. Furthermore, models which resolve vertically propagating internal gravity waves are unlikely to even include the larger scales for which this spurious momentum flux becomes significant. Notice also that this artificial increase in momentum flux is diminished as the depth of the computational domain increases. If it were necessary to remove this effect for a particular modeling application, the radiation condition can be truncated at the small-wavenumber end, where one would expect evanescent waves when $(k_x U + k_y V - \omega)^2 < f^2$. The cutoff wavenumber would be determined using a reasonable estimate for ω .

To further illustrate the performance of the simple hydrostatic radiation condition in the presence of nonhydrostatic and rotational influences, we have also computed the steady linear streamline fields for selected values of Na/U . These streamlines are obtained from the vertical displacement

$$\delta = aH \int_0^\infty \frac{[\cos lz + (\alpha^2 - 1) \cos l(z - z_i) \cos lz_i] \cos k_x x - \alpha \sin lz \sin k_x x}{\sin^2 lz_i + \alpha^2 \cos^2 lz_i} \times e^{-ak_x} dk_x, \quad (38)$$

which is derived by taking the inverse Fourier transform of (32). The solution for purely radiating waves is recovered from (38) by setting $\alpha = 1$ with the following exception: for $k < f/U$ and $k > N/U$, the α which appears unsquared in the second term of the numerator is replaced by $-i$.

The streamline fields were computed numerically from (38) with $z_i = \frac{1}{4}\lambda_z$ using both the approximate and the exact radiation conditions, and are displayed in Fig. 4 for $Na/U = 1$ and 100. At small horizontal scales (Fig. 4a), nonhydrostatic influences dominate the wave structure, while at large scales (Fig. 4b), Coriolis effects are strongly in evidence. (Compare Fig. 4 with the hydrostatic nonrotating wave structure in Fig. 1.) In both instances the approximate solution agrees very closely with the exact radiating linear solution.

These analytic results are also confirmed by linear numerical model simulations. The nonhydrostatic

model solution for $Na/U = 1$ was obtained using the compressible mountain wave model described by Durran and Klemp (1982) with the mean density variations removed. This model handles acoustic modes separately on a smaller time step than that used for the processes of physical interest. With this approach, the radiation condition (18) is evaluated only on the large time steps, even though pressure is one of the variables stepped forward in time on the small time steps. For this simulation, the large and small time steps are $U\Delta t/a = 0.04$ and $U\Delta\tau/a = 0.004$, respectively; other parameters are as in the simulations described in Section 3. Because the nonhydrostatic disturbances propagate further downstream, the downstream boundary was moved to $x = 25a$.

The evolution of the momentum flux profiles for $Na/U = 1$ are displayed in Fig. 5 for simulations using

either the radiation condition or an absorbing layer as described in reference to Fig. 1. The similarities in the development of these two model solutions verify that the approximate radiation condition has provided an accurate representation of the transient wave evolution in a situation where nonhydrostatic influences are dominant. A similar agreement was obtained for this case when the mean wind was removed and the mountain was towed along the lower boundary.

The small oscillations in the momentum flux profiles in Fig. 5 are caused primarily by the finite horizontal extent of the domain; a portion of the wave energy propagates nearly horizontally downstream and thus the vertical flux of momentum in the finite domain is not quite constant with height. In fact, comparative simulations conducted with the downstream boundary located at $x = 9a$ also produced momentum flux profiles quite similar to each other, even though the profiles displayed much more variation with height. For this case, the waves still have

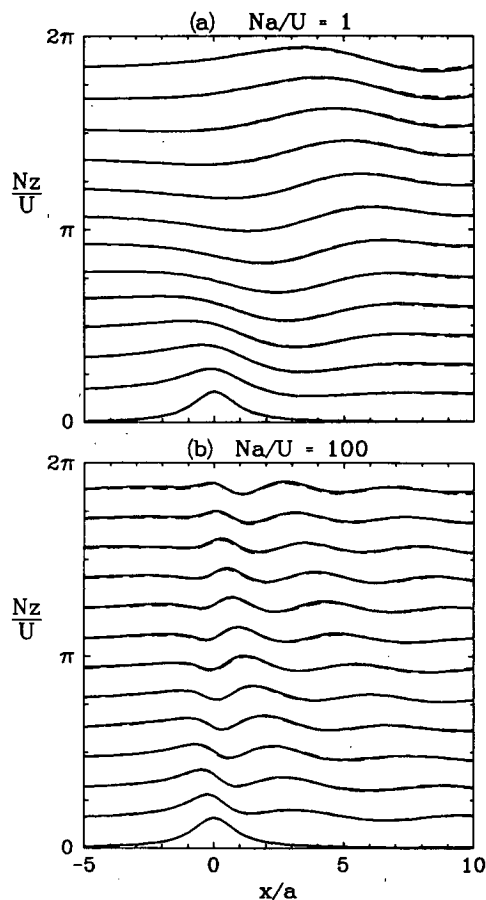


FIG. 4. Steady linear streamlines for nonhydrostatic Boussinesq flow past a bell-shaped mountain with Coriolis effects included, displayed for $NH/U = 0.5$. Solutions for the approximate (solid lines) and exact (dashed lines) radiation conditions are shown for (a) $Na/U = 1$ and (b) $Na/U = 100$, computed for $z_t = \frac{1}{4}\lambda_z$ and $f/N = 10^{-2}$.

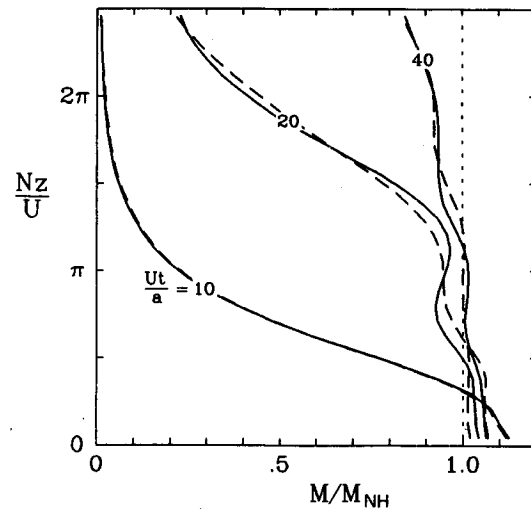


FIG. 5. Momentum flux profiles for $Ut/a = 10, 20, 40$ for linear nonhydrostatic simulations of flow over a bell-shaped mountain with $Na/U = 1$, obtained with the numerical radiation condition (solid lines) and the numerical absorbing layer (dashed lines). $M_{NH} = 0.457 M_H$ is the correct steady state momentum flux for $Na/U = 1$ (see Fig. 3).

significant amplitude at the downstream boundary (see Fig. 4a); yet the implied periodicity of the radiation condition did not significantly degrade the results.

One might expect to obtain a more accurate numerical simulation of nearly steady mountain waves by applying the radiation condition (30) with $\omega = 0$ for horizontal scales in the range $f^2 < k_x U + k_y V < N^2$. However, two-dimensional model experiments indicate that this approach creates increased error in the transient development which significantly delays the convergence to steady state. Thus, even in this application, the radiation condition (18) is the preferred approximation.

b. Nonlinear effects

In order to evaluate the ability of (18) to radiate nonlinear gravity waves, we first consider the case shown in Fig. 1 (hydrostatic, two-dimensional Boussinesq flow past a bell-shaped mountain, with Coriolis forces neglected), but now let the mountain have finite height. For constant mean wind and stability, the steady-state equations of motion reduce to a linear equation with nonlinear boundary conditions (Long, 1953), which may be solved for vertically propagating waves (cf. Huppert and Miles, 1969; Raymond, 1972; Lilly and Klemp, 1979), and used for comparison with solutions from the numerical model using the radiative upper boundary condition (18). Simulations including nonhydrostatic and Coriolis effects are discussed following this comparison.

The numerical model results for $NH/U = 0.5$ are obtained with the same model as described in Section

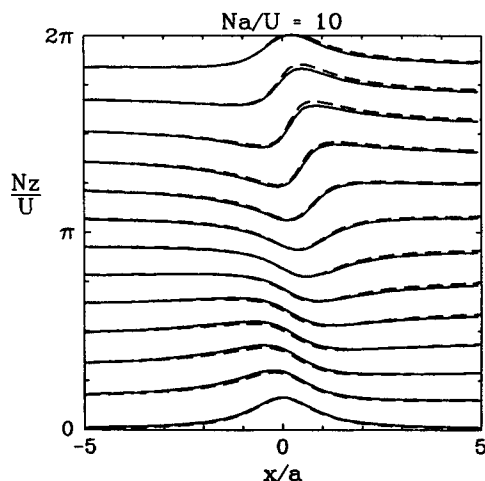


FIG. 6. As in Fig. 1, except for nonlinear simulation (solid lines) with $NH/U = 0.5$ and nonlinear solution to Long's equation (dashed lines).

3, except that the nonlinear advection terms are included in (1) and (2). The governing equations are thus given by the two-dimensional form of (24)–(28) with $f = \beta = 0$ and with (29) replaced by

$$\frac{d_H}{dt} = \frac{\partial}{\partial t} + (U + u) \frac{\partial}{\partial x} + w \frac{\partial}{\partial z}. \quad (39)$$

For these equations, the vertical advection terms in (24) and (27) require additional boundary conditions. For computational stability, we include the perturbation portion of the vertical derivative at the upper boundary only when the flow is directed out of the domain ($w > 0$) and use a one-sided difference, lagged in time. If, in more general applications, vertical mixing terms are included in the model, they should be set equal to zero along the upper boundary.

As nonlinear influences increase in the model, increased time splitting tendencies in the simulation require some additional restriction of the time step. Thus for these simulations, a dimensionless time step of $U\Delta t/a = 0.008$ was used instead of the 0.01 value used for the linear waves in Section 3. For slowly varying wave structures the simulations can be further stabilized by using a running time filter on the pressure at $z = z_i$ of the form

$$\phi(x, z_i, t + \Delta t) = r\phi^* + (1 - r)\phi(x, z_i, t), \quad (40)$$

where ϕ^* is the initial estimate of pressure obtained from the radiation condition. Using $r = 0.2$, stable simulations could again be achieved with $U\Delta t/a = 0.01$. Other numerical parameters remain the same as for the linear simulations described in Section 3.

The numerically simulated streamlines computed for $z_i = \frac{5}{4}\lambda_z$ are displayed at $Ut/a = 60$ in Fig. 6 along with the corresponding steady-state solution to Long's equation. The nonlinear analytic solution is obtained

using an iterative procedure to provide the correct radiation condition as described by Lilly and Klemm (1979). The good agreement between these two solutions suggests that (18) remains a viable approximation in this application.

The nonlinear simulations are sensitive to the depth z_i of the computational domain. Fig. 7 shows the time-dependent evolution of the pressure drag D across the mountain for simulations using the radiation condition (18) with $z_i = \frac{5}{4}\lambda_z$, $\frac{3}{2}\lambda_z$ and $\frac{7}{4}\lambda_z$, and corresponding simulations using an absorbing layer. These absorbing layer results were obtained using the same Rayleigh damping layer mentioned in Section 3, which is $1.5\lambda_z$ in depth and begins at the height $z = z_i$. The surface drag is computed by evaluating the integrated pressure difference across the mountain; for steady linear waves, the analytic solution yields $D = -M_H$. In all cases shown in Fig. 7, D rises rapidly as the simulations are initiated, reaches a maximum value of about $-1.35M_H$, and then begins to decrease toward the steady-state value of $-1.11M_H$ predicted by the nonlinear solution to Long's equation (see Lilly and Klemm, 1979). However, the surface drag then begins to drift gradually toward higher or lower values, depending on the particular value of z_i . This drift did not show strong sensitivity to the location of the lateral boundaries (moving the boundaries from $\pm 9a$ to $\pm 18a$ had little impact on the numerical solution). The simulated wave amplitudes vary correspondingly with the variations shown for surface drag.

The fact that both the approximate radiation condition and the absorbing layer exhibit this sensitivity suggests that it is not merely caused by artificial reflections at the upper boundary. Furthermore, simulations in which we doubled the absorbing layer depth (and halved the vertical gradient of the absorbing viscosity) did not noticeably diminish this sensitivity; its onset was only delayed in time. These results raise questions which cloud the interpretation of nonlinear gravity wave radiation in numerical models. Perhaps the steady-state solution to Long's equation is weakly unstable, so that small numerical disturbances cause the model solution to gradually drift to larger or smaller amplitude. Or perhaps upward propagating modes interact at various levels to produce new modes which are downward propagating. In this case, truncating the model domain at any height would eliminate internally generated reflections from above that level. Further studies designed to understand these results are being conducted, but are outside the scope of this presentation. However, we again emphasize that the nonlinear drift in the surface drag obtained using the radiation boundary condition is similar in magnitude to that generated with an absorbing layer. The nonsteady drift associated with the use of either boundary condition increases in larger amplitude mountain wave simulations.

Nonlinear, nonhydrostatic simulations were conducted for a mountain with nondimensional height $NH/U = 0.5$, but with the width decreased to $Na/U = 1$. These solutions were obtained using the same compressible nonhydrostatic mountain wave model as described in Section 4a, except that the nonlinear terms are now included. The streamline fields at $Ut/a = 60$ are shown in Fig. 8a for both the radiation condition and absorbing layer simulations (using $z_t = \frac{5}{4}\lambda_z$), and exhibit some noticeable differences. However, these differences seem to be caused by the gradual drift of the nonlinear flow field as described above in reference to Fig. 7. Earlier in the simulation, at $Ut/a = 30$, the radiation condition and absorbing layer simulations were in much closer agreement.

Nonlinear numerical simulations were also conducted for a wide mountain for which $Na/U = 100$, by including the Coriolis terms in the hydrostatic model discussed above. Again, the downstream boundary was moved to $x = 18a$ and simulations were conducted using either the radiation condition (18) or an upper absorbing layer. The resulting streamlines at $Ut/a = 60$ are displayed in Fig. 8b; the two cases are nearly identical.

c. Influence of mean density variation

Numerical models that simulate processes over a significant depth of the atmosphere must include the influence of mean density variation in the model equations. In the hydrostatic anelastic equations described by (24)–(28), this density effect arises through the βw term in the continuity equation (28). To isolate this influence on the effectiveness of the Boussinesq radiation condition (21), we consider the hydrostatic version of (24)–(28) with $f = 0$. Under these conditions, the linear deep anelastic condition for each horizontal Fourier mode at the top of the model becomes

$$\hat{\phi} = \left(\frac{N}{(k_x^2 + k_y^2)^{1/2}} + i \frac{\beta}{2} \frac{k_x U + k_y V - \omega}{(k_x^2 + k_y^2)^{1/2}} \right) \hat{w}. \quad (41)$$

The second term within the brackets in (41) represents the non-Boussinesq contribution, and complicates the radiation condition through its dependence on the frequency ω . Since this term is imaginary, it does not contribute directly to the vertical energy transport. However, omission of the term as in (21) does slightly alter the amplitudes of the vertically propagating modes from their proper values.

To illustrate the relatively minor influence of mean density variation on the radiation condition, we return to the example of steady, linear, two-dimensional, hydrostatic flow past a bell-shaped mountain with $f = 0$, but include the effects of $\beta \neq 0$. By applying the approximate radiation condition (18) at $z = z_t$, we compute the vertical momentum flux using the same procedures as those used in deriving (37),

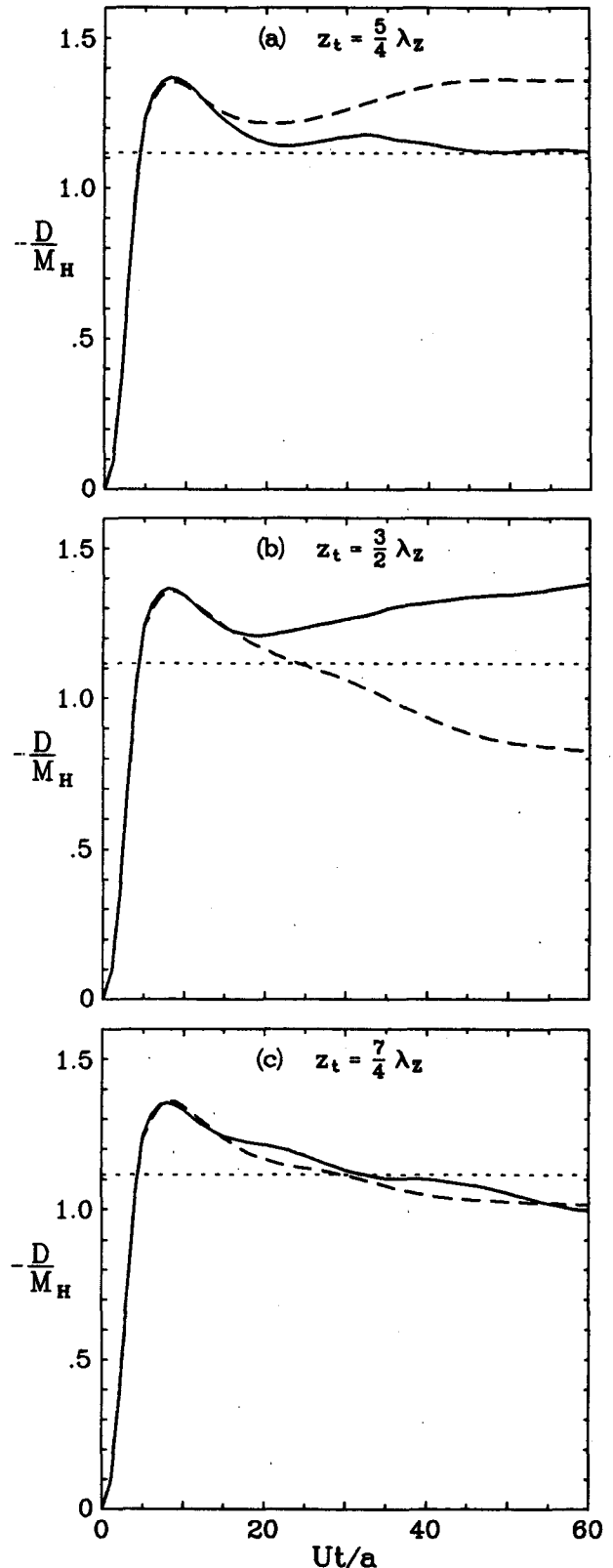


FIG. 7. Time-dependent surface drag for nonlinear simulations for $NH/U = 0.5$ using the numerical radiation condition (solid lines), or an absorbing layer of depth $1.5\lambda_z$ (dashed lines), for $z_t/\lambda_z = \frac{5}{4}, \frac{3}{2}$ and $\frac{7}{4}$.

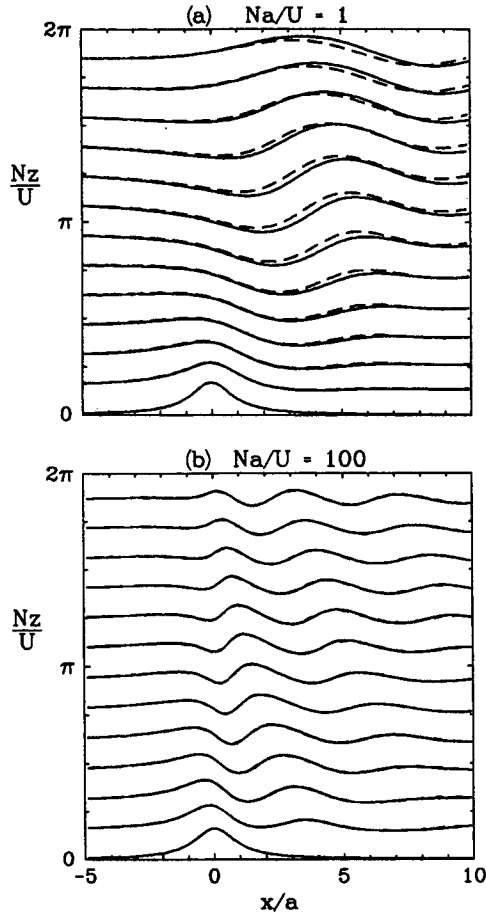


FIG. 8. Nonlinear Boussinesq streamlines at $Ua = 60$ for uniform flow over a bell-shaped mountain with $NH/U = 0.5$ using the radiating upper boundary condition (solid lines) and an upper absorbing layer (dashed lines) with $z_t = \frac{5}{4}\lambda_z$, for (a) a nonhydrostatic model with $Na/U = 1$, and (b) a hydrostatic model including Coriolis terms with $Na/U = 100$.

and compare this flux to the correct value which is again given by (23). The ratio of the approximate result to the exact solution is

$$\frac{M}{M_H} = [1 - \beta l^{-1} \sin l z_t (\cos l z_t - \beta l^{-1} \sin l z_t)]^{-1}, \quad (42)$$

where the vertical wavenumber l now satisfies

$$l^2 = \frac{N^2}{U^2} - \frac{1}{2} \frac{d\beta}{dz} - \frac{\beta^2}{4} \quad (43)$$

and is assumed to be constant in the derivation of (34). For typical atmospheric conditions $U\beta/N \ll 1$, in which case

$$\frac{M}{M_H} \approx 1 + \frac{\beta U}{2N} \sin 2 \frac{N z_t}{U} + O\left[\left(\frac{\beta U}{N}\right)^2\right]. \quad (44)$$

For example, if the radiation condition were applied

in an isothermal stratosphere where $T = 230$ K and $U = 30$ m s⁻¹, the maximum error would be

$$\frac{\beta U}{2N} = \frac{U}{2R} \left(\frac{c_p}{T}\right)^{1/2} = 0.109. \quad (45)$$

This factor becomes proportionally smaller for weaker stratospheric winds. Thus, the error in momentum flux associated with the use of the Boussinesq radiation condition in a deep atmosphere will generally be on the order of 10% or less. Note that the accuracy of this approximation also depends on the depth of the domain, since the sine factor multiplying $\beta U/2N$ in (44) oscillates between plus and minus one as z_t varies over intervals of $\frac{1}{4}\lambda_z$.

d. Effect of nonuniform atmospheric conditions

In applying the radiation upper boundary condition (21) in a numerical model we are, in effect, assuming that the atmosphere above the entire model domain is characterized by a uniform mean wind and stability. The validity of this assumption depends on the accuracy with which a constant property layer can be matched to the distribution of mean wind speed and stability along the upper boundary. If the horizontal scales of variation in the mean conditions along the top of the model domain are large compared to the scales of vertically propagating gravity waves, the proposed radiation condition should still be quite accurate. Under these conditions, (21) remains locally accurate to first order when derived using the WKB approximation. Since the horizontal wind speed does not appear in (21), gradual variations in mean winds will not significantly impair the accuracy of the radiation condition. However, variations in the mean stability along the upper boundary can cause some reflections even if the WKB approximation remains valid. If a domain-averaged stability estimate $\bar{N}$ is used in (21), reflections will be produced by vertical discontinuities in stability in regions where the local mean stability N differs from $\bar{N}$.

To illustrate the magnitude of this effect, consider again the momentum flux for steady flow past a bell-shaped mountain. Suppose a value for $\bar{N}$ is used for the stability in (18), which differs from the actual local stability N . In this case, the expression for the momentum flux becomes

$$\frac{M}{M_H} = \frac{\gamma}{\gamma^2 \sin^2(N z_t/U) + \cos^2(N z_t/U)}, \quad (46)$$

where $\gamma = \bar{N}/N$. The momentum flux again depends on z_t and is bounded such that M/M_H varies between γ and γ^{-1} . Thus, in this example, the percent error in the momentum flux is less than or equal to the percent error in estimating the local mean stability.

e. Combined influences

As a final example of the utility of this boundary condition, we compare its performance against an

absorbing layer in a more realistic mountain wave flow in which almost all the simplifying assumptions, under which (18) was derived, are violated. In this case, the atmosphere consists of a 10 km deep troposphere with a constant lapse rate of $6^{\circ}\text{C km}^{-1}$, capped by an isothermal stratosphere. The mean wind speed increases linearly from 15 to 35 m s^{-1} throughout the troposphere, and is a constant 35 m s^{-1} in the stratosphere. The mountain is bell-shaped, as described by (22), with a height of 1 km and a half width $a = 7.5\text{ km}$. The numerical solutions were generated by the full nonlinear, nonhydrostatic, compressible mountain wave model described by Durran (1981) and Durran and Klemp (1982). The finite-difference increments are chosen such that $\Delta x = 1.5\text{ km}$, $\Delta z = 333\text{ m}$, $\Delta t = 8\text{ s}$ and $\Delta \tau = \frac{8}{3}\text{ s}$ (where Δt and $\Delta \tau$ are the large and small time steps, respectively). The top of the model domain is set at $z_t = 14\text{ km}$; the lateral boundaries are at $\pm 75\text{ km}$. The absorbing layer is 16 km deep in the simulation which uses it. The mean wind is gradually initialized over a period of 2000 s.

The streamlines from the model simulation at $t = 12000\text{ s}$ are shown in Fig. 9. The simultaneous presence of a vertically propagating wave, which is basically hydrostatic, and trapped waves, which are inherently nonhydrostatic, suggest that this case is indeed more complicated than those examined earlier. Nevertheless, the solutions obtained using the radiation condition (Fig. 9a) and the wave absorbing layer (Fig. 9b) are quite similar. In addition, the surface pressure drag in the two simulations agree to within 2.5%. This comparison provides further evidence that the radiation boundary condition can be effective in practical applications.

5. Summary

A radiative upper boundary condition has been proposed for use in mesoscale models which allows most of the vertically propagating gravity wave energy to pass out of the computational domain. In this formulation, the pressure along the upper boundary is determined from the Fourier transform of the vertical velocity at that boundary. The use of this condition requires little additional computation, and has only a minimal impact on overall model efficiency. The radiation boundary condition is rigorously correct for the case of the linear, hydrostatic, Boussinesq flow in a uniform atmosphere in which Coriolis effects are ignored. We have systematically relaxed each of these assumptions, and found that the simple condition (21) remains effective in much more general atmospheric conditions, and should be useful in many mesoscale modeling applications.

The accuracy of this radiation condition has not been tested in the wide variety of modeling applications for which it is potentially useful. Instead, we have examined its behavior, in quantitative detail, for a particular application which is extremely sensitive

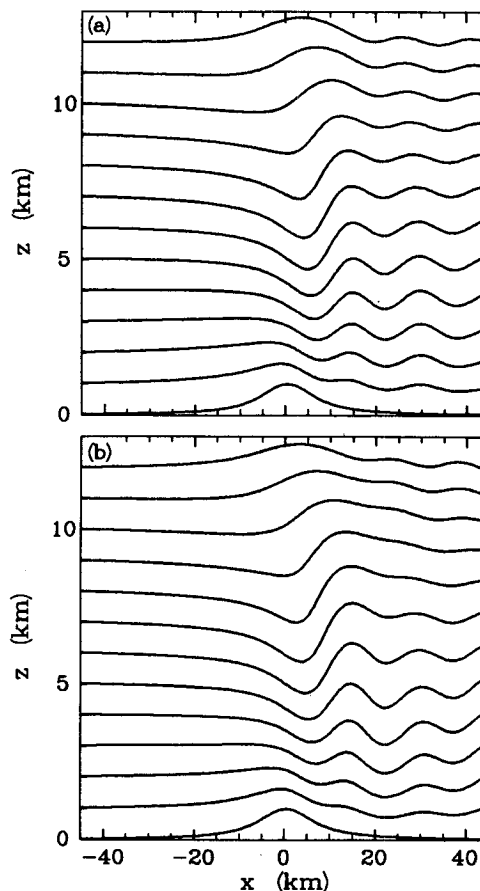


FIG. 9. Numerically computed streamlines for the mountain wave flow associated with a representative atmospheric sounding, using (a) the radiation upper boundary condition, or (b) a wave absorbing layer.

to the nature of the upper boundary condition, and for which analytic solutions are available for comparison. The radiative characteristics of both transient and steady-state two-dimensional mountain waves were analyzed to illustrate the accuracy of the approximate radiation condition when nonhydrostatic, nonlinear, non-Boussinesq and Coriolis effects are present.

Our numerical tests suggest that the radiation condition may be used in lieu of a wave absorbing layer without degrading the quality of the solution. Since the viscous absorbing layer requires an increase in the depth of the computational domain, the use of the radiation boundary condition substantially reduces both the execution time and the storage requirements of the numerical model. In the nonlinear mountain wave simulations, the wave amplitude and the associated vertical momentum transport were somewhat sensitive to the depth of the model domain; this sensitivity was encountered using either our radiation condition or an absorbing layer. These results raise questions regarding the uniqueness of nonlinear radiating gravity wave simulations which merit further investigation.

The techniques used in deriving this radiation upper boundary condition also can be applied, in principle, to the formulation of a radiating lateral boundary condition. Following this approach leads to a wave advection type boundary condition which actually can distinguish between the outward and inward propagating modes in certain simplified atmospheric environments. To illustrate this result, we again consider Eqs. (1)–(4) for an atmosphere having a uniform mean wind and stability. For vertically periodic disturbances of the form

$$\xi(x, z, t) = \hat{\xi}(x, l, t)e^{ilz}, \quad (47)$$

these equations reduce to the shallow water equations which may be expressed in the form

$$\left[\frac{\partial}{\partial t} + (U + C) \frac{\partial}{\partial x} \right] \left[\frac{\partial}{\partial t} + (U - C) \frac{\partial}{\partial x} \right] \hat{u} = 0, \quad (48)$$

where we recall that $C = N/|l|$ is positive by definition. The correct radiation condition for each vertical mode l (both positive and negative) at a lateral boundary becomes

$$\frac{\partial \hat{u}}{\partial t} + \left(U + \frac{N}{l} \right) \frac{\partial \hat{u}}{\partial x} = 0, \quad (49)$$

if $U + N/l$ is directed out through that boundary, or $\hat{u} = 0$ if $U + N/l$ is directed inward. In applying this condition to a numerical model, the normal component of horizontal velocity would be transformed vertically along the lateral boundary and also one grid point interior to the boundary in order to evaluate $\partial \hat{u} / \partial x$ using a one-sided difference. Each mode is computed at the new time level using either (49) or $\hat{u} = 0$. Finally, the inverse transform recovers the horizontal velocity at the lateral boundary. A similar approach was proposed by Pearson (1974). The primary complication in applying this procedure to realistic models is that vertical variations in the mean wind and stability are typically quite substantial, in which case the representation (47) is not appropriate for the actual wave modes. Even for uniform atmospheric conditions, a more generalized transform would be required because of the highly nonperiodic nature of the vertical wave structure. Consequently, a direct application of this approach to the lateral boundaries would require more complicated procedures than those described for the upper boundary.

The radiating upper boundary condition has not yet been tested in three-dimensional meso- β scale models where the horizontal variability in the mean conditions at the top of the domain may be significant. In these models, the extra grid points and calculations required by a wave absorbing layer impose an unacceptable computational burden, so a simple, but totally reflective, upper boundary condition usually is specified. These models are generally applied in situations in which the vertical radiation of gravity wave energy is of secondary importance. Neverthe-

less, even if the operational accuracy of the radiation condition is diminished from that documented in this paper, we believe that its use could significantly improve the treatment of gravity waves in these models.

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APPENDIX

Finite-Difference Formulation of the Radiation Boundary Condition

When the radiation boundary condition is applied in a particular model, optimum performance is obtained when (18) is derived for the particular finite-difference equations of that model. To illustrate this numerical effect, we rederive (18) for a generalized form of the finite difference equations. In (1)–(4) there are four terms containing horizontal derivatives: two of these $[\partial u / \partial x$ in (1) and $\partial \theta / \partial x$ in (2)] are associated with advection while the remaining two $[\partial \phi / \partial x$ in (1) and $\partial u / \partial x$ in (4)] are responsible for gravity wave propagation. The finite differences for these terms determine the effective representations of the horizontal wavenumbers in the system. We define the numerical equivalent of the horizontal wavenumber k_x as k_a in the finite differencing of the advection terms, and as k_w in the two x derivatives contributing to gravity wave propagation. For example, with second-order finite differencing,

$$k_a = \frac{1}{\Delta x} \sin k_x \Delta x, \quad (A1)$$

and with a standard staggered mesh (each velocity variable staggered by one-half grid interval from the location of the thermodynamic variables in the direction of the respective velocity component)

$$k_w = \frac{2}{\Delta x} \sin \frac{1}{2} k_x \Delta x, \quad (A2)$$

while for an unstaggered mesh k_w is given by (A1). (With spectral or pseudospectral models, $k_a = k_w = k_x$.) In the derivations that follow, any desired finite-difference representation can be inserted for k_a and k_w .

To derive the equivalent of (18), we replace k_x by k_a or k_w in the appropriate terms in the finite difference forms of (1)–(4), and obtain

$$\hat{\phi} = \frac{N}{|k_w|} \hat{w}. \quad (A3)$$

Notice that the numerical radiation condition depends only on the finite differencing of the gravity waves terms; it is independent of the differencing used for the advection terms.

The finite-difference representation of the radiation condition in (A3) improves the accuracy with which shorter wavelength modes are radiated in the numerical model. To illustrate these effects, consider the radiative properties of the finite-difference equations when the radiation condition is applied using an arbitrary representation k_r for the horizontal wavenumber. Solving the finite difference form of the linear wave equation (12) with the radiation condition applied at $z = z_t$, the energy flux then becomes

$$\frac{E}{E_H} = \left[1 - \left(1 - \frac{k_w^2}{k_r^2} \right) \sin^2 \left(\frac{k_w}{k_a} \frac{N z_t}{U - C_{px}} \right) \right]^{-1}, \quad (\text{A4})$$

where E_H is given by (14). When the numerical radiation boundary condition (A3) is applied, $k_r = k_w$, and thus, the correct energy flux is generated for all wavenumbers. However, when the continuous form (18) is used (i.e., $k_r = k_x$), errors arise which are functions of z_t and which grow with increasing k_x . For example, with second-order differencing on a staggered mesh, E/E_H can be as large as $\pi^2/8$ at the $4\Delta x$ wavelength, an error of about 20%.

In evaluating the numerical limitations on the time step, we consider the commonly used leapfrog time-differencing scheme. Deriving the numerical dispersion relation for (1)–(4) produces the linear stability criterion

$$\sin \omega \Delta t = \frac{U \Delta t}{\Delta x} (k_a \Delta x) \pm \frac{C \Delta t}{\Delta x} (k_w \Delta x) \leq 1, \quad (\text{A5})$$

which is independent of the upper boundary condition. Thus the time step is limited by the advection speed U and the fastest horizontally propagating gravity wave. Here C represents the finite-difference equivalent of $N/|l|$, indicating that the stability is most restricted by the smallest vertical wavenumber. Applying the radiation condition at the top of the model changes the form of the equations in the uppermost grid interval. The dispersion equation in this layer can be derived by replacing either (1) or (4) by the radiation condition (since both were used to derive it) which leads to the two roots

$$\sin \omega \Delta t = \frac{U \Delta t}{\Delta x} (k_a \Delta x) \leq 1,$$

$$\sin \omega \Delta t = \frac{U \Delta t}{\Delta x} (k_a \Delta x) - \text{sgn}(l) \frac{C \Delta t}{\Delta x} |k_r \Delta x| < 1. \quad (\text{A6})$$

For $k_r = k_w$ the stability criterion (A6) is effectively the same as (A5), i.e., the numerical radiation condition (A3) produces no additional restrictions on the linear time step criterion. For $k_r = k_x$, however, the linear stability restriction is more severe. For example, the $2\Delta x$ mode ($k_x \Delta x = \pi$) produces $|k_r \Delta x| = \pi$ if the analytic radiation condition (18) is used, while with second order differencing on a staggered grid, the numerical radiation conditions yields $|k_r \Delta x| = 2$. Since the largest C is often much greater than U , the

numerical radiation condition (A3) improves the maximum allowable linear time step by as much as 50% over that for the analytic condition (18). These differences have been confirmed for linear wave simulations. However, in nonlinear applications the increased time step restrictions discussed in Section 4 minimize the effective differences in the time step using either (18) or (A3).

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